

KL UNIVERSITY

Design and Simulation of a Digital Access Verification System Using XOR / XNOR Logic

BATCH 13

Presented by: Tanish (2620080073) • Vriithika (2620080089) • Sujeet (2620090014) • Prakash (2620080060) • Meghana (2620080049)

Agenda

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Introduction to Access Verification Systems

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Module 1 Recap — Basic Logic Gates

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XOR Gate — Working & Truth Table

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XNOR Gate — Working & Truth Table

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Bit-Comparator Concept for Access Checking

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System Block Diagram

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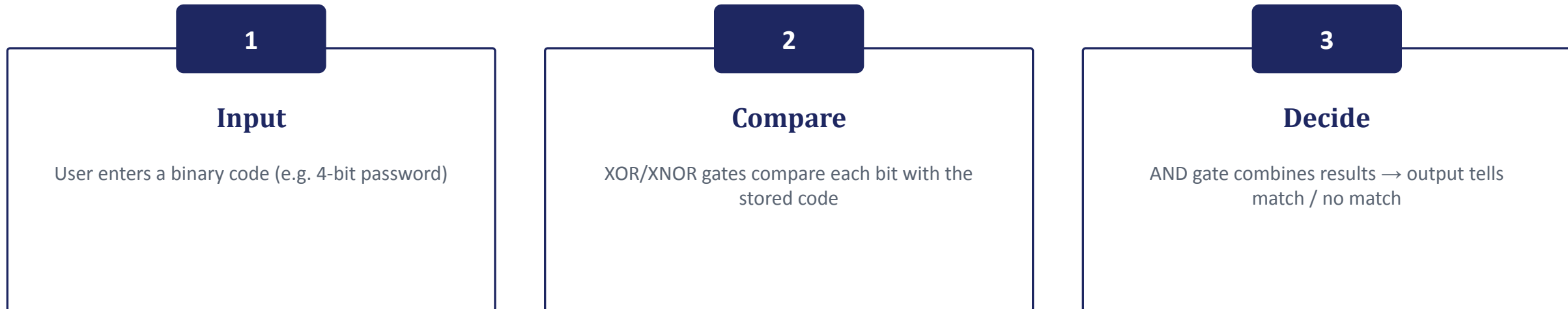
Circuit Diagram (Simulated in Logisim)

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Applications, Advantages & Conclusion

What is a Digital Access Verification System?

A circuit that checks whether an ENTERED CODE matches a STORED (authorized) CODE, and grants or denies access based on the result — the same logic used in keypad locks, ID card readers and PIN checks.



Module 1 Recap — Basic Logic Gates

AND

$$Y = A \cdot B$$

Output is 1 only if all inputs are 1

OR

$$Y = A + B$$

Output is 1 if any input is 1

NOT

$$Y = \bar{A}$$

Output is the inverse of the input

XOR

$$Y = A \oplus B$$

Output is 1 only if inputs are DIFFERENT

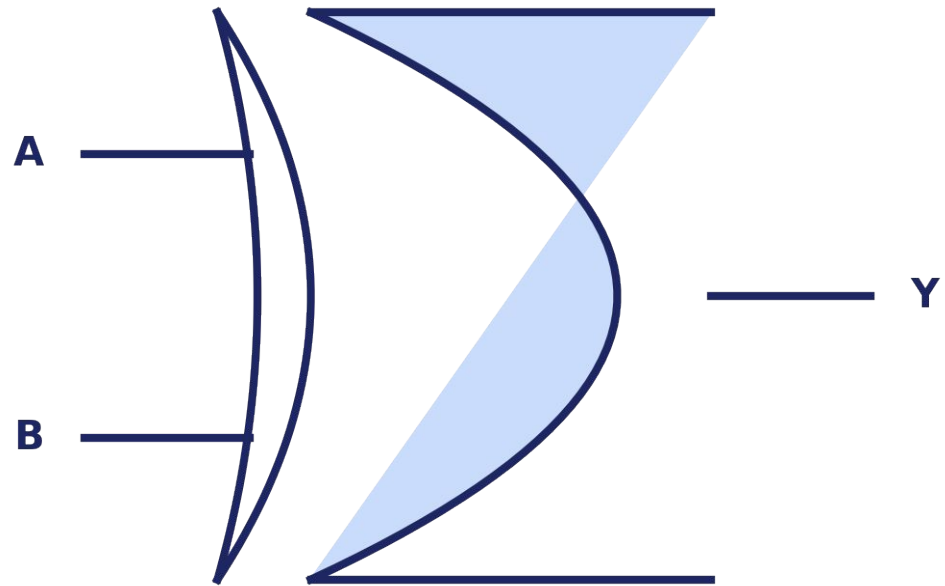
XNOR

$$Y = (A \oplus B)^\neg$$

Output is 1 only if inputs are SAME

This project focuses on XOR and XNOR — the two gates used to build the bit comparator.

XOR Gate

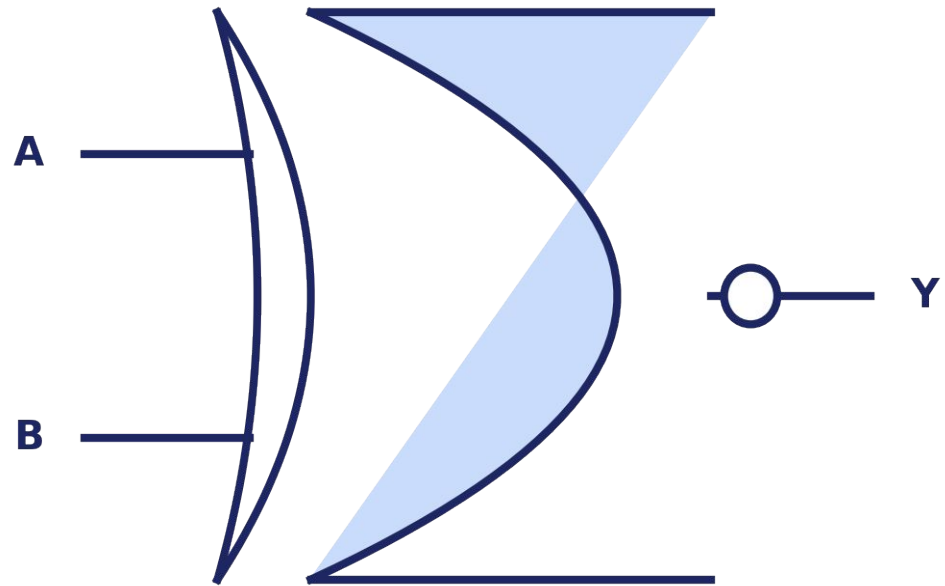


A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Boolean Expression: $Y = A \oplus B$

XOR gate can be used to identify a **mismatch** between the entered bit and the stored bit. , Output is 1 only if inputs are DIFFERENT

XNOR Gate



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Boolean Expression: $Y = (A \oplus B)'$

XNOR Gate is very useful for checking whether each bit of the entered code matches the stored code. Output is 1 only if inputs are SAME

Bit-Comparator Concept for Access Checking

Each bit of the entered code is compared with the corresponding bit of the stored code using an XNOR gate.



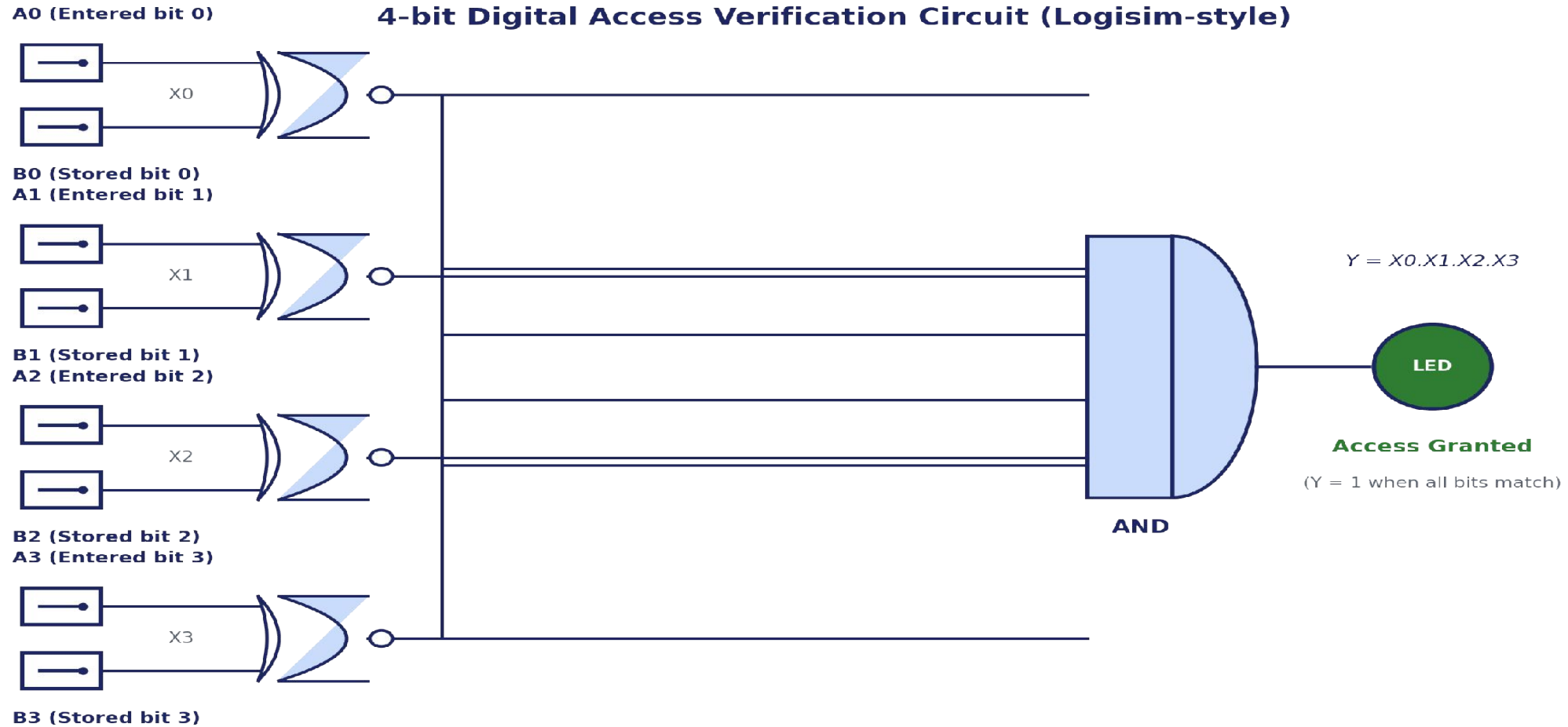
AND all XNOR outputs together → if the result = 1, EVERY bit matched → ACCESS GRANTED.
If even ONE bit differs, the AND output = 0 → ACCESS DENIED.

System Block Diagram



Two independent inputs (entered code and stored code) are compared bit-wise; the combined result drives a visible/audible output.

Circuit Diagram — Simulated in Logisim



Worked Example — 4-bit Code Check

Case 1: Correct Code Entered

Bit	A (Entered)	B (Stored)	XNOR Output
0	1	1	1
1	0	0	1
2	1	1	1
3	1	1	1

AND = 1 → ACCESS GRANTED

Case 2: Wrong Code Entered

Bit	A (Entered)	B (Stored)	XNOR Output
0	1	1	1
1	1	0	0
2	1	1	1
3	1	1	1

AND = 0 → ACCESS DENIED

Applications

1

Digital door locks & keypad entry systems

2

ATM PIN and card verification circuits

3

Password-protected embedded devices

4

Employee ID / RFID access control panels

5

Vehicle immobilizer / anti-theft systems

6

Secure locker and safe-box mechanisms

Advantages & Limitations

Advantages

- Simple, low gate-count design
- Fast comparison — works in real time
- Easy to scale to more bits (8, 16...)
- Low power and low cost to build
- Easy to simulate and test in Logisim

Limitations

- Fixed code is hardwired / not easily changed
- No encryption — code is compared in plain bits
- Vulnerable to physical tampering
- Limited to simple match / no-match logic

Conclusion

1

XOR and XNOR gates provide a simple and reliable way to compare digital codes bit by bit.

2

XNOR's “match = 1” behaviour makes it ideal for building an access/password comparator.

3

Combining XNOR outputs with a single AND gate gives one clean Grant / Deny signal.

4

The design was verified using truth tables and simulated successfully in Logisim.

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Presented By — Batch 13



Tanish | 2620080073



Vrithika | 2620080089



Sujeet | 2620090014



Prakash | 2620080060

Thank You